

Tutorial 7 – User Interface Design

1. Discuss user interface design principles.

User interface design principles are essential for creating effective and user-friendly interfaces. These principles include:

- **Consistency:** Ensuring that similar elements behave in similar ways to create a predictable environment for users.
- **Feedback:** Providing users with clear and immediate responses to their actions to help them understand the results of their interactions.
- **Simplicity:** Keeping the interface simple and uncluttered to make it easy for users to navigate and find what they need.
- **Visibility:** Making important elements visible and easily accessible to users.
- **Affordance:** Designing elements in a way that suggests their functionality, making it intuitive for users to interact with them.

2. What is the classification of user interaction styles?

User interaction styles can be classified into several categories, including:

- **Command Line Interface (CLI):** Users interact with the system by typing commands.
- **Menu-Driven Interface:** Users select options from menus.
- **Graphical User Interface (GUI):** Users interact with the system through graphical elements like buttons and icons.
- **Form-Fill Interface:** Users fill out forms to input data.
- **Natural Language Interface:** Users interact with the system using natural language, either spoken or written.

3. Discuss three advantages of graphical information display and suggest three applications where it would be more appropriate to use graphical rather than digital displays of numeric information.

- Able to convey information effectively. Eg. Using graph to show analysis result is more efficient than using word and numeric figure for explanation.
- Avoid any misunderstanding on word. Using graphic is clearer.
- With different colour applied in graphical information, it highlight the importance of specific information.
- Eg,. Application are : Graph analysis software, Map, process description software(Scientific)

4. What are the guidelines that should be followed when using colours in a user interface? Suggest how colour might be used more effectively in the interface of a customer self-ordering system in a restaurant.

When using colours in a user interface, it's important to follow certain guidelines to ensure the interface is both aesthetically pleasing and functional. Here are some key guidelines:

- **Consistency:** Use a consistent colour scheme throughout the interface to create a cohesive look and feel.
- **Contrast:** Ensure there is sufficient contrast between text and background colours to make the text readable.
- **Accessibility:** Choose colours that are accessible to users with colour vision deficiencies. Tools like colour contrast checkers can help ensure your choices meet accessibility standards.
- **Meaningful Use:** Use colours to convey meaning and provide visual cues. For example, red can indicate errors or warnings, while green can indicate success or confirmation.
- **Limit Colour Palette:** Avoid using too many colours, which can make the interface look cluttered and confusing. Stick to a limited palette to maintain visual harmony.
- **Cultural Considerations:** Be aware of cultural differences in colour perception and meaning. Colours can have different connotations in different cultures.

In the context of a customer self-ordering system in a restaurant, colour can be used more effectively in the following ways:

- **Highlighting Options:** Use different colours to highlight different categories of food (e.g., appetizers, main courses, desserts) to make it easier for customers to navigate the menu.
- **Feedback:** Use colours to provide immediate feedback on actions. For example, use green to indicate that an item has been successfully added to the order, and red to indicate an error or unavailable item.
- **Promotions and Specials:** Use bright, attention-grabbing colours to highlight special offers or promotions, making them stand out to customers.
- **Navigation:** Use colours to guide customers through the ordering process. For example, use a different colour for the "Next" button to make it clear how to proceed to the next step.

5. Suppose you are the online shoppers for lazada.com.my **Online Shopping Portal System**.

"The design of human-computer interaction for lazada.com is heavily influenced by non-functional requirements."

Do you agree with the above statement? Give **TWO (2)** reasons to explain your answer.

Yes. For example, usability, the interface should provide support to users on the access of system's interface. Learnability, able to learn the system within short period. These are non-functional requirements.



Additional Questions

- a) A development team is developing a system which is used for furniture design. The system should provide interface to the junior designer to draft and apply notation from the palette provided by system. Senior designer uses the system to fine tune the draft and instruct the system to perform necessary tasks with special instruction type into the system. They are less depending on the graphical interface.
- (i) Suggestion **TWO (2)** user interaction styles for the above scenario. Provide justification for your suggestions. (6 marks)
- Menu style
Command Style
- (ii) Analyse **TWO (2)** challenges you may face when designing the interface for the above system. (4 marks)

- To embed the considerations for different levels of designer.
- Mixture of design led to more complication and more factors to be considered.

(iii) Analyse **TWO (2)** usability attributes that are important for the system. (6 marks)

Refer to notes.

(iv) Discuss **TWO (2)** ways of evaluating the system interface design. (4 marks)

- Perform usability test by users.
- Organize review on the user interface.